Software Project Management – Project Description

Project Title:

CampusConnect: Unified Event Portal for TMU University

Project Overview:

TMU University seeks to boost student engagement by developing **CampusConnect**, a centralized web platform for managing and promoting all campus events. This system will allow students to discover events, RSVP, and track their participation. For student clubs and university departments, it will provide tools to create, manage, and analyze their events.

The application will consist of a student-facing event discovery portal and an organizer-facing management dashboard. The project is broken down into eight sub-projects, each delivering a core piece of functionality.

Key Features and Sub-Project Breakdown:

1. User Authentication & Role Management

- **Team:** Identity & Access Team
- **Key Features:**
  - **University Single Sign-On (SSO) Integration:** Students and staff log in with their existing university credentials.
  - **Role-Based Permissions:** Defines permissions for Students, Club Leaders, Department Admins, and System Administrators.
  - **Club Registration:** Allows student groups to register as official event organizers on the platform.
- **Integration:**
  - The foundation of the system. Provides user identity and roles to all other sub-projects.

2. Event Discovery & Public Calendar

- **Team:** UX & Front-End Team
- **Key Features:**
  - **Browse & Filter Events:** Students can view events by date, category (academic, social, sports), location, and sponsoring organization.
  - **Search Functionality:** Search for events by keywords in the title or description.
  - **Integrated Calendar View:** View events in a daily, weekly, or monthly calendar format.
- **Integration:**
  - Pulls all event data from the **Event Management backend (Sub-project 3)**.
  - Displays data from the **User Dashboard (Sub-project 4)** like personal RSVPs.

3. Event Creation & Management System (Organizer-Facing)

- **Team:** Core Services Team
- **Key Features:**
    - **Event Creation Form:** Organizers can create events with details like title, description, date, time, location, and capacity.
    - **Edit & Cancel Events:** Organizers can modify event details or cancel events, triggering notifications to attendees.
    - **Attendance Tracking:** Provides a check-in list or QR code for organizers to track who attended their event.
- **Integration:**
    - Directly feeds the **Event Discovery calendar (Sub-project 2)**.
    - Connects to the **RSVP System (Sub-project 5)** for attendee lists.

4. User Personal Dashboard

- **Team:** User Engagement Team
- **Key Features:**
    - **My Events:** A personalized view of events the user has RSVP'd to, is interested in, or has attended.
    - **Event Recommendations:** Suggests events based on the user's interests and past attendance.
    - **Achievements & Points:** Dispoints earned for participating in events (see Sub-project 6).
- **Integration:**
    - Central hub for user-specific data, pulling from **RSVPs (5)**, **Rewards (6)**, and **Notifications (8)**.

5. RSVP & Interest Management System

- **Team:** Logistics Team
- **Key Features:**
    - **RSVP Workflow:** Students can confirm attendance for events with limited capacity.
    - **"Interest" Feature:** For events without capacity limits, users can mark themselves as "interested" to get a reminder.
    - **Waitlist Management:** Automatically manages a waitlist for full events and promotes waitlisted users if spots open up.
- **Integration:**
    - Critical link between **Users (1)**, **Events (3)**, and **Notifications (8)**.

6. Student Engagement & Rewards Module

- **Team:** Gamification Team
- **Key Features:**
    - **Points System:** Awards points to students for attending events, especially those from different categories.
    - **Rewards Catalog:** Allows students to redeem points for rewards (e.g., university merch, coffee shop discount codes).
    - **Leaderboard:** Optional feature to show the most engaged students or clubs.
- **Integration:**
    - Tracks attendance from the **Event Management system (3)**.
    - Redeems points through the **User Dashboard (4)**.

7. Organizer Analytics & Reporting Dashboard

- **Team:** Data & Analytics Team
- **Key Features:**
    - **Event Performance Reports:** Provides organizers with data on RSVPs, actual attendance, and attendance rate for their events.
    - **Feedback Summary:** Aggregates star ratings and comments from attendees (see Sub-project 8).
    - **Audience Insights:** Shows the colleges and majors of students who attended.
- **Integration:**
    - Pulls data from the **RSVP (5)**, **Event Management (3)**, and **Feedback (8)** systems.
    - Accessed by organizers through their view of the system.

8. Event Feedback & Notification System

- **Team:** Comms Team
- **Key Features:**
    - **Post-Event Surveys:** Prompts attendees to rate an event and provide optional feedback.
    - **Automated Notifications:** Sends emails for event reminders, waitlist updates, canceled events, and new reward opportunities.
    - **Broadcast Announcements:** Allows university admins to send announcements to all users or specific groups.
- **Integration:**
    - A utility service used by nearly all other sub-projects (e.g., **RSVP (5)**, **Events (3)**, **Rewards (6)**) to communicate with users.

9. Room & Resource Booking System

- **Team:** Facilities & Logistics Team
- **Key Features:**
    - **Room Availability Check:** Organizers can view and book available classrooms, lecture halls, and meeting rooms for their events directly through the platform.
    - **Resource Request:** Request additional resources like projectors, microphones, or catering services linked to the event booking.
    - **Approval Workflow:** Sends booking requests to facility managers for official confirmation and scheduling.
- **Integration:**
    - **Deeply integrated with Event Creation:** The **Event Creation System (Sub-project 3)** must be modified to include a room booking step during the event setup process.
    - Provides data to the **Public Calendar (Sub-project 2)** to show event location details.
    - Triggers **Notifications (Sub-project 8)** for booking confirmations, denials, and reminders to facility managers.

Timeline and Sub-Project Phases:

- **Week 1-2:** Project Planning & Team Setup
- **Week 3-4:** Requirements Gathering & System Design
- **Week 5-8:** Core Development (Sprints for each sub-project)
- **Week 9:** Integration & Testing Week
- **Week 10:** Final Demo, Presentation, and Submission

Evaluation:

- **Project Planning & Management (40%):** Quality of the project plan, timeline (Gantt chart), risk assessment, and team coordination.
- **Sub-project Functionality (30%):** Each sub-project is demoed and graded on whether it meets its core requirements.
- **Integration and Usability (10%):** How well the sub-projects work together as one system. Is the user journey smooth?
- **Documentation & Final Presentation (20%):** Quality of technical documentation, user manuals, and the final team presentation.